

NLP HW3 — Q2: WhoDunnit (Multi-Agent Systems)

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2.1 Naive Agent Implementation (Analysis)

We ran the naive baseline with the default configuration and no additional flags (`python run_game.py --iterations 3`: 3 suspects, `max_turn_count = 11`, no summarizer, no internal thought). *Alex* is the Accuser and *Beth* the Intel agent. The full log is in Appendix ??; the analysis below cites it directly.

What specific behavioral problems did you identify in the initial run of the naive agents? Provide at least two concrete examples from your `game_log`.

Problem A — Analysis paralysis: the Accuser will not accuse even when the culprit is already uniquely determined.

- **Game 1** (culprit = Suspect 3): only Suspect 3 wears square glasses, so Beth's answer `square eye glasses` → `Suspect 3` already fixed the culprit. Alex still asked four more broad questions and never accused, hitting the turn limit.
- **Game 2** (culprit = Suspect 3): `blue eye color` → S2, S3 and `circular eye glasses` → S1, S3 intersect to {S3}, yet Alex kept questioning until timeout.

Problem B — No cross-turn reasoning. The Accuser treats each answer in isolation and never intersects the narrowing suspect lists into a running candidate set, so it cannot recognize that only one suspect remains — the root cause of Problem A.

Do the agents dynamically utilize the full spectrum of their available action space, or do they prematurely converge on a single, repetitive action type? Analyze how this behavioral pattern affects their overall progress.

They **converge**. Alex issues only `request-broad` in all three games (zero `request-specific`; `accuse` only once, in Game 3), which in turn forces Beth into `respond-broad` only. Because the terminal `accuse` action is under-used, information accumulates but is rarely acted upon, so games stall: Games 1 and 2 exhaust all 11 turns without an accusation, ending in timeout rather than a decision.

Does the Accuser make a blind accusation prematurely, or does it get stuck in an endless loop until the `max-turn-count` is reached? Why do you think it struggles to finalize the decision?

It gets stuck in an **endless questioning loop**, not a premature accusation: Games 1 and 2 reach `max-turn-count` with no accusation, and Game 3 accuses correctly only after redundant confirmation. It struggles to finalize because:

1. **Stateless prompting** — no maintained candidate set, so it never sees that one suspect remains;
2. the naive prompt **forbids reasoning**, so the model defaults to asking one more question rather than committing;

3. **no firm stopping criterion**, so “accuse when certain” never triggers.

2.2 Context Management via Summarization

We implemented `SummerizerAgent.summarize` as an evidence-preserving LLM rewrite of the channel (`temperature=0.2`, prompt instructed to keep every per-suspect confirmed/excluded attribute and never invent facts) and ran with the summarizer enabled (`python run_game.py --include-summerizer --iterations 3`). The Summarizer is *Charlie*. The full log is in Appendix ??.

Did the summarizer inadvertently remove any critical clues that led to the Accuser failing to identify the culprit? Provide a specific example from the logs.

The decisive clues were generally *kept*, but the summarizer distorted evidentiary *certainty*. In Game 1 Beth stated definitely that **circular eye glasses** → **Suspect 1, Suspect 2, Suspect 3**, yet Charlie rewrote this as “Suspect 1: *possibly* has circular eye glasses” (and likewise for S2, S3), downgrading a confirmed fact to an uncertain one. This ambiguity plausibly caused Alex to redundantly re-ask both “blue eye color” and “circular eye glasses,” exhausting its turns without accusing — even though the culprit-identifying clue (only Suspect 3 has blue eyes) had actually been preserved.

Did the reduction in context length result in more coherent questions responses from the Accuser and Intel agents, or did the loss of “raw” dialogue history make their behavior less natural?

Mixed, but coherence improved in 2 of 3 games. The per-suspect organized summaries (e.g. Game 2: “Suspect 2: blue eye color, wears square eye glasses, has long hair”) kept the Accuser tightly focused on the correct suspect and pushed it toward targeted **request-specific** questions — an improvement over the naive baseline’s endless broad queries. But when a summary was vague (Game 1’s “possibly has”), the loss of the verbatim Q&A record removed Alex’s memory of what it had already asked and it repeated questions. So compression helps when the summary is faithful and structured, and hurts when it introduces ambiguity.

Based on your observations, do you believe this summarization approach would scale to a game with a much larger suspect pool (e.g., suspect-count=20)? Explain your reasoning.

Not without changes. (1) A free-text LLM rewrite is lossy: the “possibly” distortion seen at 3 suspects will compound at 20, and a single weakened or dropped fact can leave the culprit ambiguous. (2) The summary must record facts for many suspects, so its length grows with the pool and the “concise channel” benefit erodes. (3) The real bottleneck is the Accuser’s reasoning and stopping ability, which summarization does not address — and a third of all turns are consumed by summarizing. A deterministic, structured state (a maintained per-suspect table of confirmed/excluded attributes) would scale far better than a natural-language summary.

A Naive baseline game log

```
Game 1
Suspects:
Suspect 1: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a black
           hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
           have long, black pants and a pink, button-up shirt.
Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown
           hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they
           have short, black pants and a pink, button-up shirt.
```

Suspect 3: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a silver watch, they have long, black pants and a green, tee shirt.

==

The culprit is: 3

==

Communication channel:

Alex asked a broad question about: green eye color

Beth responded with: Suspect 2, Suspect 3

Alex asked a broad question about: square eye glasses

Beth responded with: Suspect 3

Alex asked a broad question about: long hair

Beth responded with: Suspect 1, Suspect 2, Suspect 3

Alex asked a broad question about: black hat

Beth responded with: Suspect 1, Suspect 3

Alex asked a broad question about: red shoes

Beth responded with: Suspect 2, Suspect 3

Alex asked a broad question about: silver watch

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Game 2

Suspects:

Suspect 1: their eye color is green, they have circular eye glasses, their hair is short, is wearing a black hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they have short, black pants and a pink, tee shirt.

Suspect 2: their eye color is blue, they have square eye glasses, their hair is short, is wearing a brown hat, their hobby is dancing, is in a happy mood, they have red shoes, they have a silver watch, they have short, black pants and a green, tee shirt.

Suspect 3: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a silver watch, they have short, brown pants and a green, tee shirt.

==

The culprit is: 3

==

Communication channel:

Alex asked a broad question about: blue eye color

Beth responded with: Suspect 2, Suspect 3

Alex asked a broad question about: circular eye glasses

Beth responded with: Suspect 1, Suspect 3

Alex asked a broad question about: short hair

Beth responded with: Suspect 1, Suspect 2, Suspect 3

Alex asked a broad question about: brown hat

Beth responded with: Suspect 2, Suspect 3

Alex asked a broad question about: white shoes

Beth responded with: Suspect 1, Suspect 3

Alex asked a broad question about: silver watch

=====

Game 3

Suspects:

Suspect 1: their eye color is brown, they have square eye glasses, their hair is short, is wearing a black hat, their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they have short, black pants and a green, button-up shirt.

Suspect 2: their eye color is green, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a bronze watch, they have short, black pants and a pink, button-up shirt.

Suspect 3: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a bronze watch, they have short, black pants and a pink, tee shirt.

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The culprit is: 1

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Communication channel:

Alex asked a broad question about: brown eye color

Beth responded with: Suspect 1, Suspect 3

Alex asked a broad question about: square eye glasses

Beth responded with: Suspect 1

Alex asked a broad question about: short hair

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Beth responded with: Suspect 1
Alex asked a broad question about: black hat
Beth responded with: Suspect 1
Alex accused Suspect 1 and was correct!
```

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```

B Summarizer-enabled game log

```
Game 1
Suspects:
Suspect 1: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a brown
          hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a bronze watch, they
          have long, brown pants and a pink, tee shirt.
Suspect 2: their eye color is brown, they have circular eye glasses, their hair is long, is wearing a black
          hat, their hobby is dancing, is in a sad mood, they have red shoes, they have a silver watch, they have
          short, black pants and a green, tee shirt.
Suspect 3: their eye color is blue, they have circular eye glasses, their hair is short, is wearing a brown
          hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a bronze watch, they
          have long, black pants and a pink, tee shirt.
==
The culprit is: 3
==
Communication channel:
Alex asked a broad question about: blue eye color
Beth responded with: Suspect 3
Charlie summarized the game so far: Alex asked about blue eye color and Beth responded that Suspect 3 has it.

Alex asked a broad question about: circular eye glasses
Beth responded with: Suspect 1, Suspect 2, Suspect 3
Charlie summarized the game so far: Suspect 1: possibly has circular eye glasses.
Suspect 2: possibly has circular eye glasses.
Suspect 3: has blue eye color, possibly has circular eye glasses.

Alex asked a broad question about: blue eye color
Beth responded with: Suspect 3
Charlie summarized the game so far: Suspect 1: possibly has circular eye glasses.
Suspect 2: possibly has circular eye glasses.
Suspect 3: has blue eye color, possibly has circular eye glasses.
Alex asked about blue eye color and Beth responded with Suspect 3.

Alex asked a broad question about: circular eye glasses
Beth responded with: Suspect 1, Suspect 2, Suspect 3

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Game 2
Suspects:
Suspect 1: their eye color is brown, they have circular eye glasses, their hair is short, is wearing a brown
          hat, their hobby is basketball, is in a happy mood, they have red shoes, they have a silver watch, they
          have long, black pants and a green, tee shirt.
Suspect 2: their eye color is blue, they have square eye glasses, their hair is long, is wearing a brown hat,
          their hobby is dancing, is in a happy mood, they have red shoes, they have a bronze watch, they have
          short, black pants and a green, button-up shirt.
Suspect 3: their eye color is brown, they have square eye glasses, their hair is long, is wearing a brown hat,
          their hobby is basketball, is in a happy mood, they have white shoes, they have a silver watch, they
          have short, brown pants and a green, button-up shirt.
==
The culprit is: 2
==
Communication channel:
Alex asked a broad question about: blue eye color
Beth responded with: Suspect 2
Charlie summarized the game so far: Suspect 2 has blue eye color.

Alex asked a specific question: Does Suspect 2 wear square eye glasses?
```

Beth responded with: Yes
 Charlie summarized the game so far: Suspect 2: blue eye color, wears square eye glasses.
 Alex asked if Suspect 2 wears square eye glasses, Beth answered yes.

Alex asked a specific question: Does Suspect 2 have long hair?
 Beth responded with: Yes
 Charlie summarized the game so far: Suspect 2: blue eye color, wears square eye glasses, has long hair.
 Alex asked if Suspect 2 wears square eye glasses, Beth answered yes.
 Alex asked if Suspect 2 has long hair, Beth answered yes.

Alex asked a specific question: Does Suspect 2 wear a brown hat?
 Beth responded with: Yes

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Game 3
 Suspects:
 Suspect 1: their eye color is green, they have square eye glasses, their hair is long, is wearing a black hat, their hobby is basketball, is in a happy mood, they have white shoes, they have a silver watch, they have short, black pants and a pink, button-up shirt.
 Suspect 2: their eye color is blue, they have circular eye glasses, their hair is long, is wearing a brown hat, their hobby is basketball, is in a sad mood, they have red shoes, they have a bronze watch, they have long, black pants and a green, tee shirt.
 Suspect 3: their eye color is brown, they have square eye glasses, their hair is short, is wearing a brown hat, their hobby is dancing, is in a happy mood, they have white shoes, they have a bronze watch, they have long, black pants and a green, tee shirt.

==

The culprit is: 2

==

Communication channel:
 Alex asked a broad question about: blue eye color
 Beth responded with: Suspect 2
 Charlie summarized the game so far: Suspect 2 has blue eye color.

Alex asked a specific question: Does Suspect 2 wear circular eye glasses?
 Beth responded with: Yes
 Charlie summarized the game so far: Suspect 2: blue eye color, wears circular eye glasses.
 Alex asked if Suspect 2 wears circular eye glasses, Beth answered yes.

Alex asked a specific question: Does Suspect 2 have long hair?
 Beth responded with: Yes
 Charlie summarized the game so far: Suspect 2: blue eye color, wears circular eye glasses, has long hair.
 Alex asked if Suspect 2 wears circular eye glasses, Beth answered yes.
 Alex asked if Suspect 2 has long hair, Beth answered yes.

Alex asked a specific question: Does Suspect 2 wear a brown hat
 Beth responded with: Yes

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